

Michael Ichikawa

AI ENGINEER · ML ENGINEER · DATA SCIENTIST

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SUMMARY

I shipped a production AI agent for Microsoft 365 solo, from empty repo to live and transactable on the Microsoft Commercial Marketplace. It runs a plan/act/observe loop over 85 function-calling tools across 12 services, with KNN tool retrieval, RAG memory, and an eval framework that gated every release. Alongside it I built a multi-agent workflow engine from scratch. Before that, MS Mathematics, eight years teaching undergraduate math, and three years of semiconductor physical design at Intel.

FLAGSHIP WORK

Microclaw — Production AI Agent for Microsoft 365

Live on the Microsoft Marketplace

A single AI agent operating across 12 Microsoft 365 services through 85 function-calling tools. Iterative plan/act/observe orchestration loop (Azure OpenAI, function-calling), two-stage KNN tool retrieval (95.0% recall vs a 92.7% embedding-only baseline, benchmarked at 1,020 examples), RAG semantic memory, smart model routing, and multi-layer failure recovery.

TypeScript · Azure OpenAI · Microsoft Graph · Azure Postgres (RLS) · Bicep

Multi-Agent Workflow Engine — built from scratch

mtichikawa.github.io/workflow-engine

A Python engine that turns a plain-English request into a working multi-agent workflow, composed from 14 modular, single-purpose agents with human approval before anything acts. It reuses existing agents instead of near-duplicating them, and I tested it on real public data in 12 unrelated domains, keeping weaker results marked provisional. Bidirectional MCP interop, calling tools on external MCP servers and publishing its own workflows as MCP tools, added as an edge-only adapter with no core changes and real protocol-level tests.

Python · agentic orchestration · MCP (client + server) · contracts + recipes · static graph validation

TECHNICAL SKILLS

Agentic Dev	Claude Code (daily driver) · agent-directed builds verified by tests and evals · per-repo context files · agent co-authored commit history
LLM / Agents	agent orchestration · multi-agent workflows · MCP (Model Context Protocol) · KNN tool retrieval · RAG · Azure OpenAI · Anthropic API · Microsoft Graph · HuggingFace · FinBERT
Testing / Eval	eval frameworks · A/B against baselines · property-based testing · differential testing · mutation testing · LLM-as-judge · pytest · Vitest · GitHub Actions CI
Databases / Data	PostgreSQL (row-level security) · SQLite · BigQuery · Databricks · Delta Lake · Spark Structured Streaming · dbt · SQLAlchemy · pandas
ML / Statistics	scikit-learn · PyTorch · Prophet · XGBoost · LightGBM · scipy · statsmodels · Bayesian inference
Cloud / Infra	Azure (Container Apps · Postgres · Entra · Bicep) · AWS (S3 · Lambda · DynamoDB) · GCP (Vertex AI · Cloud Run) · Docker · FastAPI (REST) · Redis
Languages	Python · SQL · TypeScript · Node.js · C++ · R

EXPERIENCE

Founding AI/ML Engineer

2026 – present

Microclaw LLC · Portland, OR

- Took the agent above from an empty repo to live and transactable on the Microsoft Commercial Marketplace in 11 weeks, solo, and passed Microsoft AppSource certification (Standard + Self-Hosted offerings)
- Built the commerce integration: Marketplace Fulfillment API v2 and the Metering Service API for per-message metered billing, Microsoft as merchant of record, across eight pricing tiers
- Multi-tenant isolation via Postgres row-level security: FORCE policies on every tenant-scoped table, an app role that cannot bypass them, delegated Microsoft Entra auth, and org plus per-user permission layers

Mathematics Faculty

2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught undergraduate mathematics for eight years: Algebra through Calculus III, Linear Algebra, Differential Equations, Statistics
- Designed assessments and iterated course materials from cohort performance data
- Taught STEM majors alongside non-STEM majors fulfilling general-education math requirements at an open-enrollment community college

Physical Design Engineer

2008 – 2011

Intel Corporation · Semiconductor R&D

- Microprocessor physical design on Intel Ivy Bridge (22nm), including timing closure, power analysis, and design rule verification
- Built Python and Perl automation for physical verification workflows, processing millions of layout data points per run

EDUCATION

MS Mathematics

The City College of New York

2014

BS Mechanical Engineering

University of California, Berkeley

2008

PORTFOLIO PROJECTS — full portfolio at mtichikawa.github.io

Self-directed work in data and ML, beyond the two flagships above. **Trading system arc** — a paper trading pipeline from exchange data through charting and signal fusion to a backtester and Streamlit dashboard (live demo). **Databricks medallion lakehouse** on NYC taxi data. **RAG pipeline** on Vertex AI and BigQuery Vector Search. **Streaming analytics** on Spark and Redpanda.

Also: real-time anomaly detection · cloud ETL on AWS · SQL analytics pipeline · LLM data assistant · multi-armed bandit A/B testing · LLM bias detection · financial NLP parser · GitHub trend forecaster · C++/pybind11 rolling stats.